

MULTI-LLM BENCHMARKING & EVALUATION PLATFORM

Technical Documentation & API Reference Manual

1. SYSTEM OVERVIEW & ARCHITECTURE

The Multi-LLM Benchmarking & Evaluation Platform is an enterprise-grade Django REST API application designed to dynamically process user prompts, evaluate AI model outputs through a double-blind jury system, and resolve a definitive winner utilizing a statistical consensus-scoring mechanism.

Core Architectural Pipeline:

1. **Ingress Validation:** The client submits a payload containing a prompt. The platform filters and validates the input length via serialization constraints.
2. **Asynchronous Orchestration:** Using Python's `asyncio` engine, the backend dispatches non-blocking concurrent API requests to multiple upstream LLM providers (OpenAI, Gemini, DeepSeek, Groq).
3. **Double-Blind Anonymization:** Model identities are stripped dynamically and mapped to uppercase token labels (`Response A`, `Response B`, etc.) to guarantee unbiased downstream evaluation.
4. **Independent Evaluation Jury:** Anonymized outputs are parsed through multi-agent AI Judges (Gemini and Groq) evaluating core linguistic vectors: Accuracy, Relevance, Completeness, Clarity, and Conciseness.
5. **Statistical Consensus Blending :** Text structures undergo gemini vectorization and Cosine Similarity checking to isolate aberrant hallucinations or outliers, updating model scores mathematically before crowning the session winner.
6. **Relational Persistence:** Complete session audit logs are atomically saved inside a relational storage engine (PostgreSQL) using optimized query techniques.

2. ENVIRONMENT CONFIGURATION (`.env`)

To run this application locally or in production, create a `.env` file in the root project directory. Copy the sample block below and replace the placeholders with your actual API gateway credentials.

```
# Server Deployment Configuration
```

```
DEBUG=True
```

```
SECRET_KEY=django-insecure-mock-key-evaluation-platform-2026
```

Upstream LLM Provider API Keys

OPENROUTERAPI_KEY=sk-proj-mockOpenAiKeyForTestingPurposesOnly12345

GEMINI_API_KEY=AlzaSyMockGeminiKeyGenAIPlatformRegistration2026

GROQ_API_KEY=gsk_mockGroqHighPerformanceInferenceEngineKey99

Database Configuration (Uncomment for PostgreSQL Setup)

DB_NAME=multi_llm_bench_db

DB_USER=postgres

DB_PASSWORD=your_secure_db_password

DB_HOST=localhost

DB_PORT=5432

3. API SPECIFICATIONS & DOCUMENTATION

ENDPOINT 1: Submit Prompt & Execute Evaluation Benchmark

- **URL:** `/api/evaluate/`
- **Method:** `POST`
- **Headers:** `Content-Type: application/json`

http://127.0.0.1:8000/api/evaluate/

Server response

Code	Details
200	Response body

```
{
  "session_id": 1,
  "prompt": "what is blackhole",
  "winner": "Openrouter-gpt-oss-20b:free",
  "final_score": 9.5,
  "all_responses": [
    {
      "label": "A",
      "model_name": "Groq-llama-3.1-8b-instant",
      "response_text": "A black hole is a region in space where the gravitational pull is so strong that nothing, including light, can escape. It is formed when a massive star collapses in on itself and its gravity becomes so strong that it warps the fabric of spacetime around it.\n\nHere's a simplified explanation of how a black hole is formed:\n\n1. **Massive Star Collapse**: A massive star runs out of fuel and dies. Its core collapses under its own gravity, causing a massive amount of matter to be compressed into an incredibly small space.\n\n2. **Singularity Formation**: The collapse creates a point of infinite density and zero volume, known as a singularity. This is the center of the black hole.\n\n3. **Event Horizon Formation**: As matter falls towards the singularity, it creates a boundary called the event horizon. Once something crosses the event horizon, it is trapped by the black hole's gravity and cannot escape.\n\n4. **Gravitational Pull**: The event horizon marks the point of no return. Any matter or radiation that enters the event horizon is pulled towards the singularity, never to escape.\n\n**Characteristics of Black Holes**:\n\n1. **Mass**: Black holes have mass, which determines the strength of their gravity.\n\n2. **Spin**: Black holes can rotate, which affects the direction of their jets of energy and the way they interact with their surroundings.\n\n3. **Charge**: Black holes can have an electric charge, which affects their behavior in the presence of other charged objects.\n\n**Types of Black Holes**:\n\n1. **Stellar Black Holes**: Formed from the collapse of individual stars, these black holes have masses up to about 20 times that of the sun.\n\n2. **Supermassive Black Holes**: Found at the centers of galaxies, these black holes have masses millions or even billions of times that of the sun.\n\n3. **Intermediate-Mass Black Holes**: Black holes with masses between those of stellar and supermassive black holes.\n\n**Observational Evidence for Black Holes**:\n\n1. **X-rays and Gamma Rays**: Telescopes can detect X-rays and gamma rays emitted by hot gas swirling around black holes.\n\n2. **Radio Waves**: Radio telescopes can detect radio waves emitted by matter as it spirals into a black hole.\n\n3. **Gravitational Waves**: The detection of gravitational waves by LIGO and VIRGO provide strong evidence for the existence of black holes.\n\n**Interesting Facts about Black Holes**:\n\n1. **Information Paradox**: The laws of quantum mechanics suggest that information cannot be destroyed, but the laws of general relativity suggest that information is lost in a black hole. This paradox is still an open question in physics.\n\n2. **Black Hole Entropy**: Black holes have a temperature and entropy, which is related to the area of their event horizon.\n\n3. **Black Hole Complementarity**: The information paradox is resolved by considering that information that falls into a black hole is not lost, but is instead stored on the event horizon and can be retrieved in a different form."
    }
  ]
}
```

Response headers

Without addition of consensus

POST <http://127.0.0.1:8000/api/evaluate/> Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings Cookies

Headers 8 hidden

Key	Value	Description	Bulk Edit	Presets	...
Content-Type	application/json				

Body Cookies Headers 10 Test Results 200 OK • 15.41 s • 5.05 KB

JSON Preview Visualize

```
1 {
2   "session_id": 2,
3   "prompt": "briefly tell what is machine learning?",
4   "winner": "Groq-llama-3.1-8b-instant",
5   "final_score": 9.0,
6   "all_responses": [
7     {
8       "label": "A",
9       "model_name": "Groq-llama-3.1-8b-instant",
10      "response_text": "**Machine Learning (ML)** is a subset of artificial intelligence (AI) that involves training algorithms",
11      "response_time_ms": 1085.81,
12      "groq_score": 9,
13      "gemini_score": 9.0,
14      "final_score": 9.0
15    },
16    {
17      "label": "B",
18      "model_name": "Groq-qwen/qwen3-32b",
```

With consensus

POST <http://127.0.0.1:8000/api/evaluate/> Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

Body Cookies Headers 10 Test Results 200 OK • 34.13 s • 10.55 KB

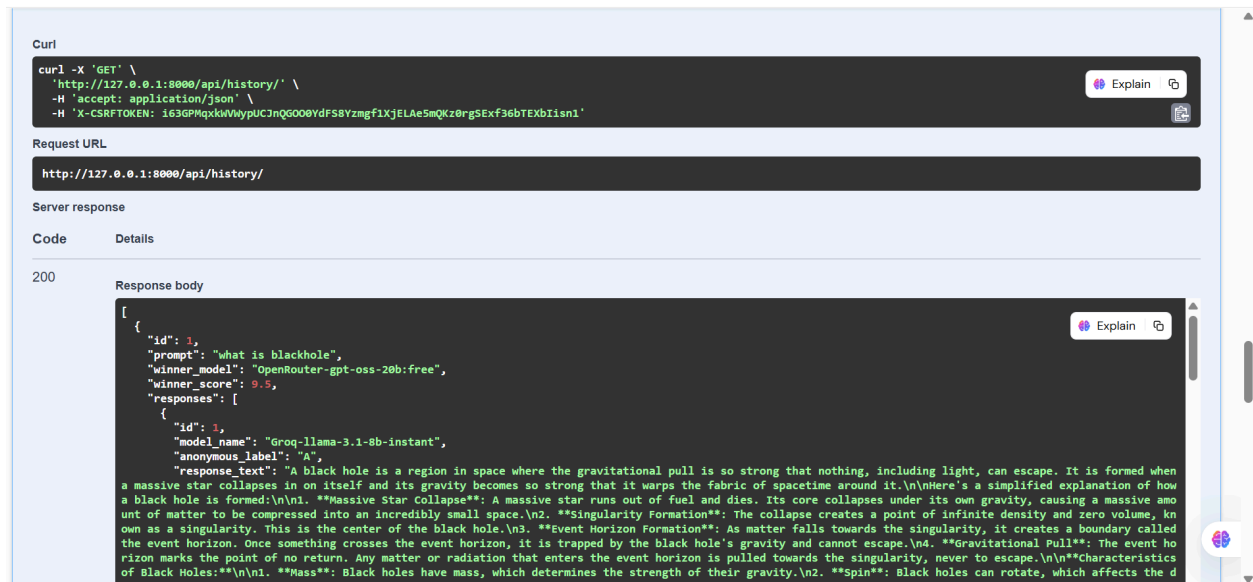
JSON Preview Visualize

```
1 {
2   "session_id": 6,
3   "prompt": "AI KYA HA?",
4   "winner": "OpenRouter-gpt-oss-20b:free",
5   "final_score": 6.98,
6   "judges_used": {
7     "groq": true,
8     "gemini": true
9   },
10  "consensus": {
11    "scores": {
12      "A": 0.0954,
13      "B": 0.1083,
14      "C": 0.1022,
15      "D": 0.0916
16    },
17    "outliers": [],
18    "confidence": "low",
```

ENDPOINT 2: Historical Evaluation Sessions Log

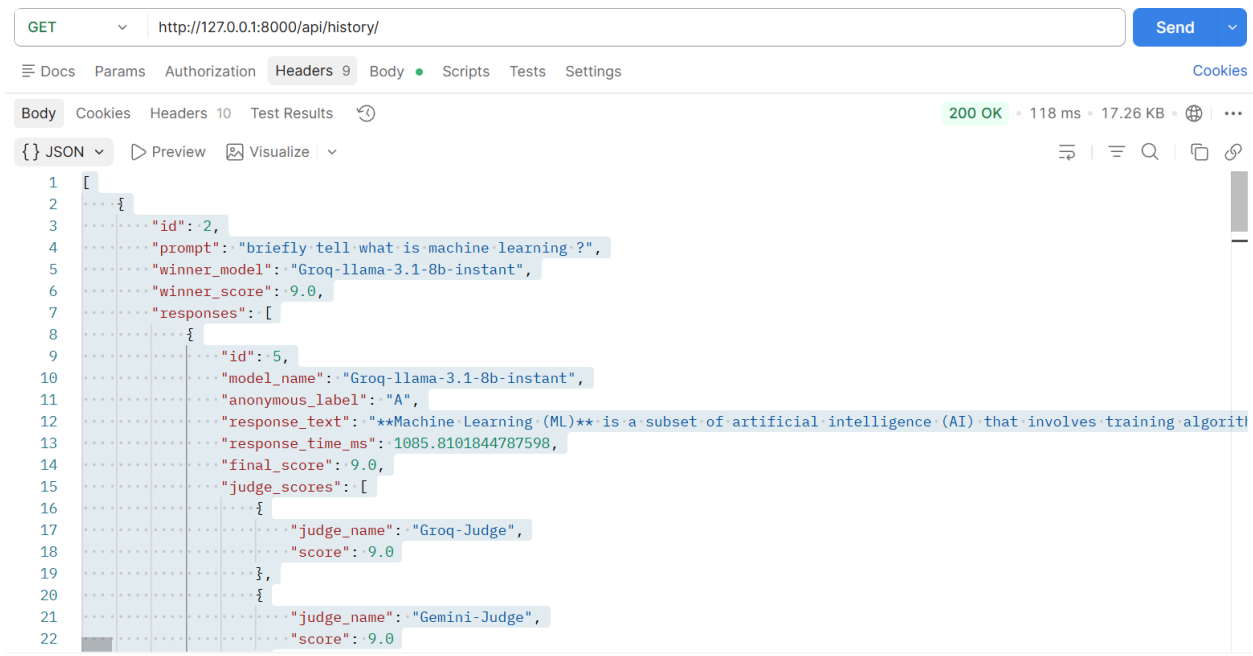
- URL: </api/history/>
- Method: GET
- Headers: Accept: [application/json](#)

- **Description:** Retrieves a registry containing all historical model comparison evaluation records. The ORM query layers execute structural `prefetch_related` wrappers to eliminate database performance bottlenecks (N+1 query loops).



The screenshot shows a web browser window with a REST client interface. The 'Curl' tab is active, displaying a curl command to fetch data from `http://127.0.0.1:8000/api/history/`. The 'Request URL' is also set to this endpoint. The 'Server response' tab shows a 200 status code. The 'Response body' is a JSON array containing a single object with the following structure:

```
{
  "id": 1,
  "prompt": "what is blackhole",
  "winner_model": "OpenRouter-gpt-oss-20b-free",
  "winner_score": 9.5,
  "responses": [
    {
      "id": 1,
      "model_name": "Groq-llama-3.1-8b-instant",
      "anonymous_label": "A",
      "response_text": "A black hole is a region in space where the gravitational pull is so strong that nothing, including light, can escape. It is formed when a massive star collapses in on itself and its gravity becomes so strong that it warps the fabric of spacetime around it. Here's a simplified explanation of how a black hole is formed:\n\n1. **Massive Star Collapse**: A massive star runs out of fuel and dies. Its core collapses under its own gravity, causing a massive amount of matter to be compressed into an incredibly small space.\n\n2. **Singularity Formation**: The collapse creates a point of infinite density and zero volume, known as a singularity. This is the center of the black hole.\n\n3. **Event Horizon Formation**: As matter falls towards the singularity, it creates a boundary called the event horizon. Once something crosses the event horizon, it is trapped by the black hole's gravity and cannot escape.\n\n4. **Gravitational Pull**: The event horizon marks the point of no return. Any matter or radiation that enters the event horizon is pulled towards the singularity, never to escape.\n\n**Characteristics of Black Holes**:\n\n1. **Mass**: Black holes have mass, which determines the strength of their gravity.\n\n2. **Spin**: Black holes can rotate, which affects the structure of their event horizon and the paths of objects nearby.\n\n3. **Temperature**: Black holes are not completely black; they emit a faint glow known as Hawking radiation, though it is extremely difficult to detect."
    }
  ]
}
```



The screenshot shows a REST client interface with a GET request to `http://127.0.0.1:8000/api/history/`. The response is a 200 OK status with a response time of 118 ms and a body size of 17.26 KB. The response body is a JSON array containing a single object with the following structure:

```
{
  "id": 2,
  "prompt": "briefly tell what is machine learning?",
  "winner_model": "Groq-llama-3.1-8b-instant",
  "winner_score": 9.0,
  "responses": [
    {
      "id": 5,
      "model_name": "Groq-llama-3.1-8b-instant",
      "anonymous_label": "A",
      "response_text": "**Machine Learning (ML)** is a subset of artificial intelligence (AI) that involves training algorithms to learn from data and make predictions or decisions based on that learning.",
      "response_time_ms": 1085.8101844787598,
      "final_score": 9.0,
      "judge_scores": [
        {
          "judge_name": "Groq-Judge",
          "score": 9.0
        },
        {
          "judge_name": "Gemini-Judge",
          "score": 9.0
        }
      ]
    }
  ]
}
```

ENDPOINT 3: Isolated Session Details

- **URL:** `/api/history/<id>/` (e.g., `/api/history/1/`)
- **Method:** GET
- **Headers:** Accept: application/json

- **Description:** Pulls isolated deep audit logs for a single target execution session parsed via its integer primary key ID. Validates route data types, rendering automatic **404 Not Found** messages if indices do not exist.

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/history/' \
  -H 'accept: application/json' \
  -H 'X-CSRFToken: 1636PMqxkMwYpUCJnQG00YdFSBYzmGf1XjELae5mQKz0rg5Exf36bTeXbI1sn1'
```

Request URL

```
http://127.0.0.1:8000/api/history/
```

Server response

Code
Details

200

Response body

```
{
  "id": 1,
  "prompt": "what is blackhole",
  "winner_model": "OpenRouter-gpt-oss-20b-free",
  "winner_score": 9.5,
  "responses": [
    {
      "id": 1,
      "model_name": "Groq-llama-3.1-8b-instant",
      "anonymous_label": "A",
      "response_text": "A black hole is a region in space where the gravitational pull is so strong that nothing, including light, can escape. It is formed when a massive star collapses in on itself and its gravity becomes so strong that it warps the fabric of spacetime around it. Here's a simplified explanation of how a black hole is formed:\n\n1. **Massive Star collapse**: A massive star runs out of fuel and dies. Its core collapses under its own gravity, causing a massive amount of matter to be compressed into an incredibly small space.\n2. **Singularity Formation**: The collapse creates a point of infinite density and zero volume, known as a singularity. This is the center of the black hole.\n3. **Event Horizon Formation**: As matter falls towards the singularity, it creates a boundary called the event horizon. Once something crosses the event horizon, it is trapped by the black hole's gravity and cannot escape.\n4. **Gravitational Pull**: The event horizon marks the point of no return. Any matter or radiation that enters the event horizon is pulled towards the singularity, never to escape.\n\n**Characteristics of Black Holes**:\n\n1. **Mass**: Black holes have mass, which determines the strength of their gravity.\n2. **Spin**: Black holes can rotate, which affects the distribution of their mass and energy.\n3. **Charge**: Black holes can have an electric charge, which affects their behavior."

```

GET
http://127.0.0.1:8000/api/history/2
Send

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Body Cookies Headers 10 Test Results

200 OK • 93 ms • 5.63 KB

{} JSON Preview Visualize

```

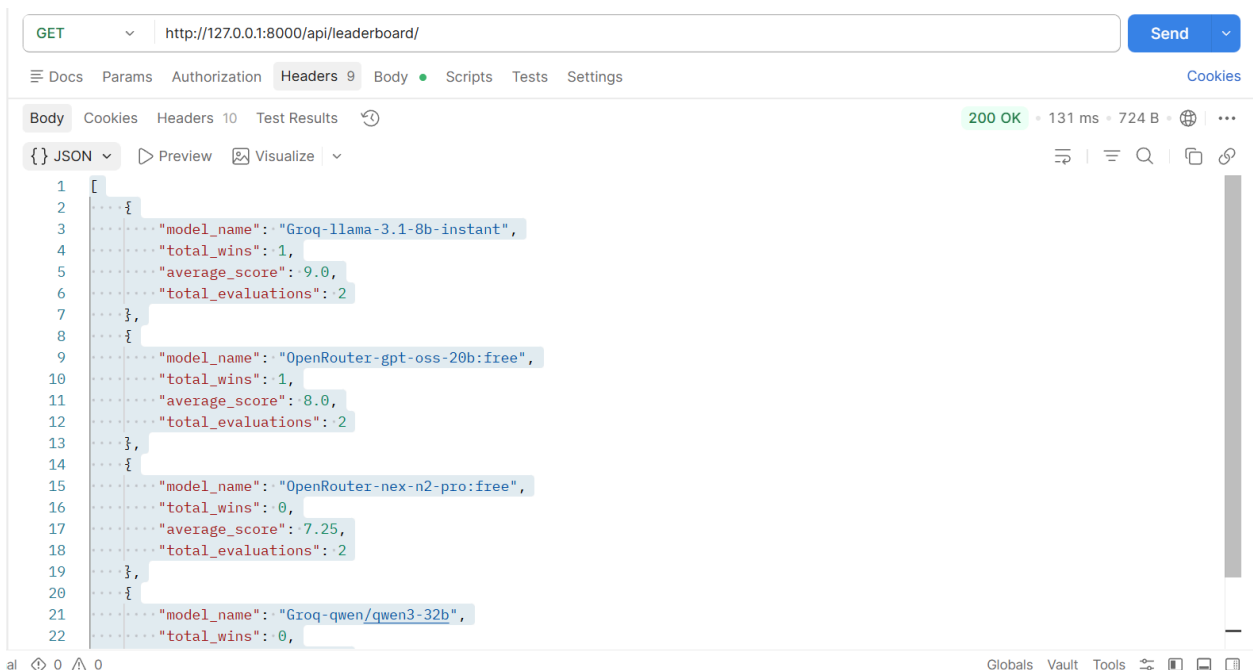
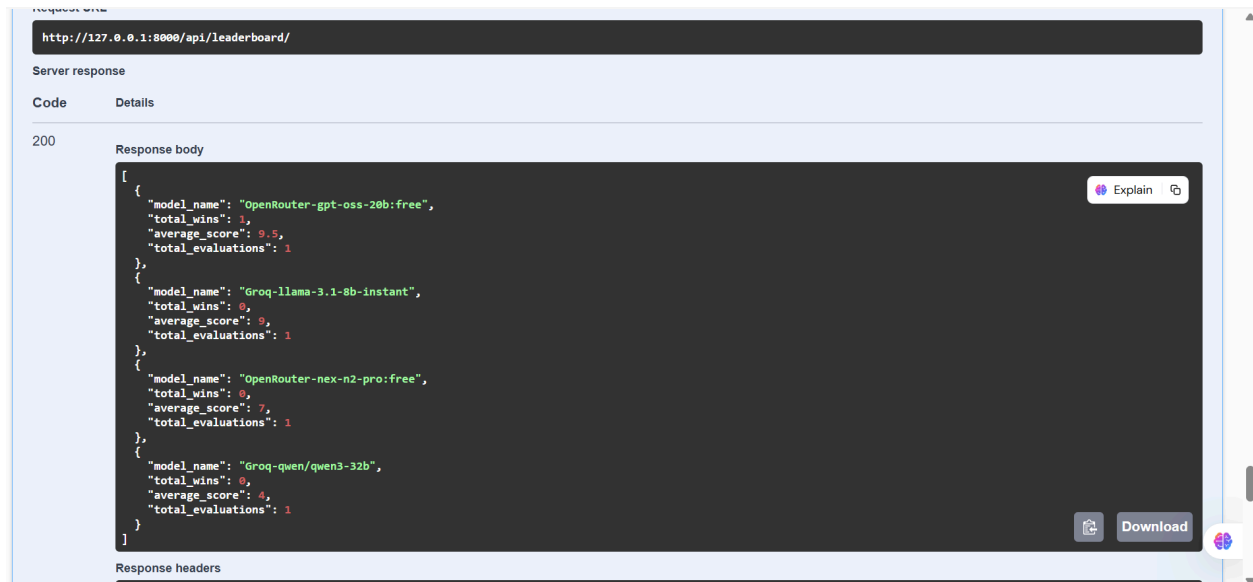
1 {
2   "id": 2,
3   "prompt": "briefly tell what is machine learning?",
4   "winner_model": "Groq-llama-3.1-8b-instant",
5   "winner_score": 9.0,
6   "responses": [
7     {
8       "id": 5,
9       "model_name": "Groq-llama-3.1-8b-instant",
10      "anonymous_label": "A",
11      "response_text": "**Machine Learning (ML)** is a subset of artificial intelligence (AI) that involves training algorithms",
12      "response_time_ms": 1085.8101844787598,
13      "final_score": 9.0,
14      "judge_scores": [
15        {
16          "judge_name": "Groq-Judge",
17          "score": 9.0
18        },
19        {
20          "judge_name": "Gemini-Judge",
21          "score": 9.0
22        }
23      ]
24    }
25  ]
26 }

```

ENDPOINT 4: Global Performance Leaderboard Rankings

- **URL:** `/api/leaderboard/`
- **Method:** `GET`
- **Headers:** `Accept: application/json`

- Description:** Compiles real-time analytical metrics by grouping historical evaluation blocks at the database layer. Leverages Django **Count** and **Avg** functions to compute global model performance stats, ordering the output dynamically using total match wins.



4. ADVANCED FEATURE OVERVIEW: AI-DRIVEN STATISTICAL CONSENSUS ENGINE

To meet the evaluation Bonus Challenge criteria, the platform implements an embedded mathematical tracking ecosystem inside **consensus.py** instead of relying blindly on simple judge score averages.

- **Semantic Similarity Vectoring:** Generated model response texts are parsed dynamically using **google/gemini-embedding-2** via OpenRouter. Instead of just counting words, the system converts text into high-dimensional vectors to capture deep semantic meaning. It then calculates cross-response agreement distributions using **Cosine Similarity** checks (1.0 = identical meaning, 0.0 = unrelated).
- **Algorithmic Outlier Isolation:** Any model response displaying extreme deviations from structural similarity averages (Standard Deviation boundary factors exceeding $1.5 \times \sigma$ combined with absolute raw similarity margins falling below 0.15) is flagged as an aberrant outlier or potential hallucination.
- **Weighted Score Penalty Mitigation:** When an outlier node is isolated, its consensus factor is penalized (cut by 50%) and dynamically down-weighted to minimize its chance of being selected as the winner, preventing corrupted or off-topic responses from gaming the leaderboard rankings.

$$\text{Adjusted Score} = (\text{Raw Judge Score} \times (1 - w)) + (\text{Consensus Score} \times 10 \times w)$$

Where:

- $w = 0.20$ (Consensus weight = 20%)
- $1 - w = 0.80$ (Judge weight = 80%)

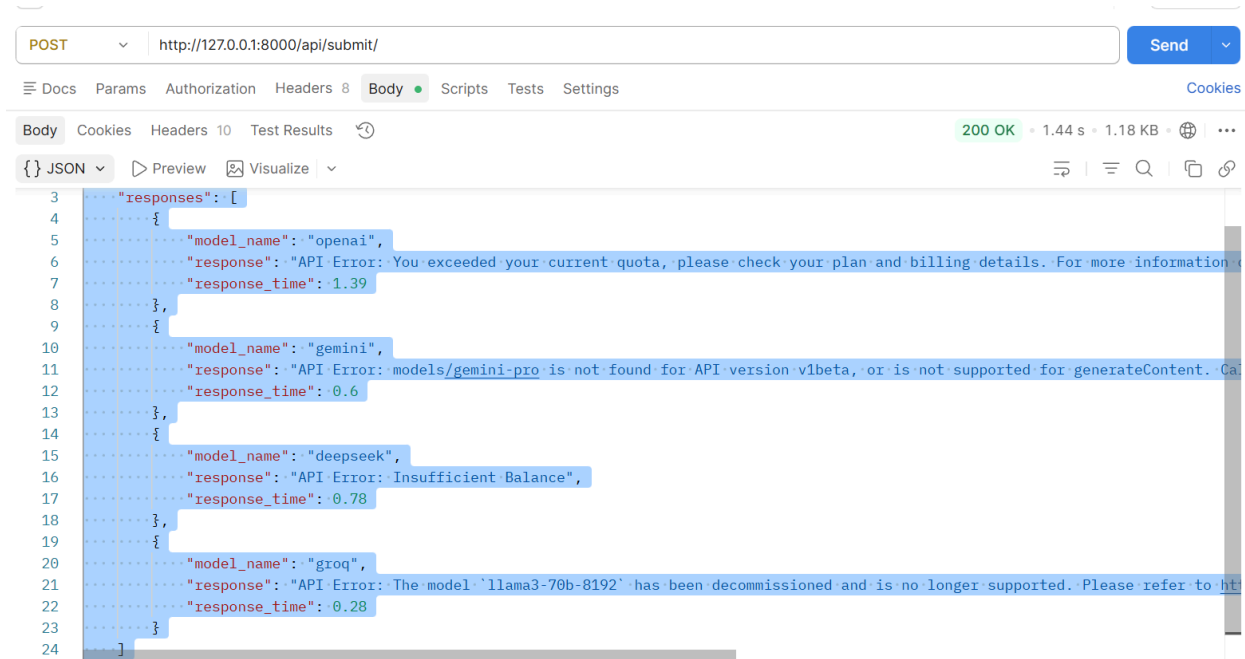
Challenges Faced and Resolutions

1. OpenAI API Quota Exceeded

Issue: You exceeded your current quota, please check your plan and billing details.

Cause: The API key was valid, but the OpenAI account had no available quota or active billing.

Resolution: OpenAI integration done using openrouter api.



```
POST http://127.0.0.1:8000/api/submit/
200 OK • 1.44 s • 1.18 KB
Body
JSON
[
  {
    "model_name": "openai",
    "response": "API Error: You exceeded your current quota, please check your plan and billing details. For more information",
    "response_time": 1.39
  },
  {
    "model_name": "gemini",
    "response": "API Error: models/gemini-pro is not found for API version v1beta, or is not supported for generateContent. Ca",
    "response_time": 0.6
  },
  {
    "model_name": "deepseek",
    "response": "API Error: Insufficient Balance",
    "response_time": 0.78
  },
  {
    "model_name": "groq",
    "response": "API Error: The model 'llama3-70b-8192' has been decommissioned and is no longer supported. Please refer to ht",
    "response_time": 0.28
  }
]
```

2. Gemini Model Not Found

Issue:models/gemini-1.5-flash is not found for API version v1b

Cause:The application was using an outdated Gemini model that was no longer available for the provided API version.

Resolution:Available models were retrieved using the Gemini Models API and the integration was updated to use gemini-2.5-flash

3. Gemini Free Tier Quota Exceeded

Issue:Quota exceeded for metric generate_content_free_tier_requests

Cause:the Gemini free-tier request limit had been reached.

Resolution:Retry handling was considered and alternative judge models were used when quota limits were encountered.

4. DeepSeek Insufficient Balance

POST

https://api.deepseek.com/chat/completions

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Params

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Body

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Tests

Se

Query Params

Key	Value
Key	Value

Body

Cookies

Headers 14

Test Results

{ } JSON

Preview

Debug with AI

```
1 {
2   "error": {
3     "message": "Insufficient Balance",
4     "type": "unknown_error",
5     "param": null,
6     "code": "invalid_request_error"
7   }
8 }
```

https://api.deepseek.com/chat/completions

Save

Share

GET

https://api.deepseek.com/chat/completions

Send

Docs

Params

Authorization

Headers 9

Body

Scripts

Tests

Settings

Cookies

☐ none

☐ form-data

☐ x-www-form-urlencoded

☒ raw

☐ binary

☐ GraphQL

JSON

Schema

Beautify

```
6   "content": "Explain Django ORM in simple words"
7   }
8 ]
9 }
```

Body

Cookies

Headers 9

Test Results

403 Forbidden • 1.08 s • 1.23 KB •

HTML

Preview

Try with different auth credentials

0 0 0

Globals Vault Tools

Issue:Insufficient Balance

Cause:The DeepSeek account did not have sufficient credits to process requests.

Resolution:Resolved by using api from openrouter provider .

5. Groq Model Deprecation

Issue:The model llama-3.1-70b-versatile has been decommissioned

Cause:The selected Groq model had been officially deprecated.

Resolution:The deprecated model was replaced with supported models:

llama-3.1-8b-instant

qwen/qwen3-32b

openai/gpt-oss-20b

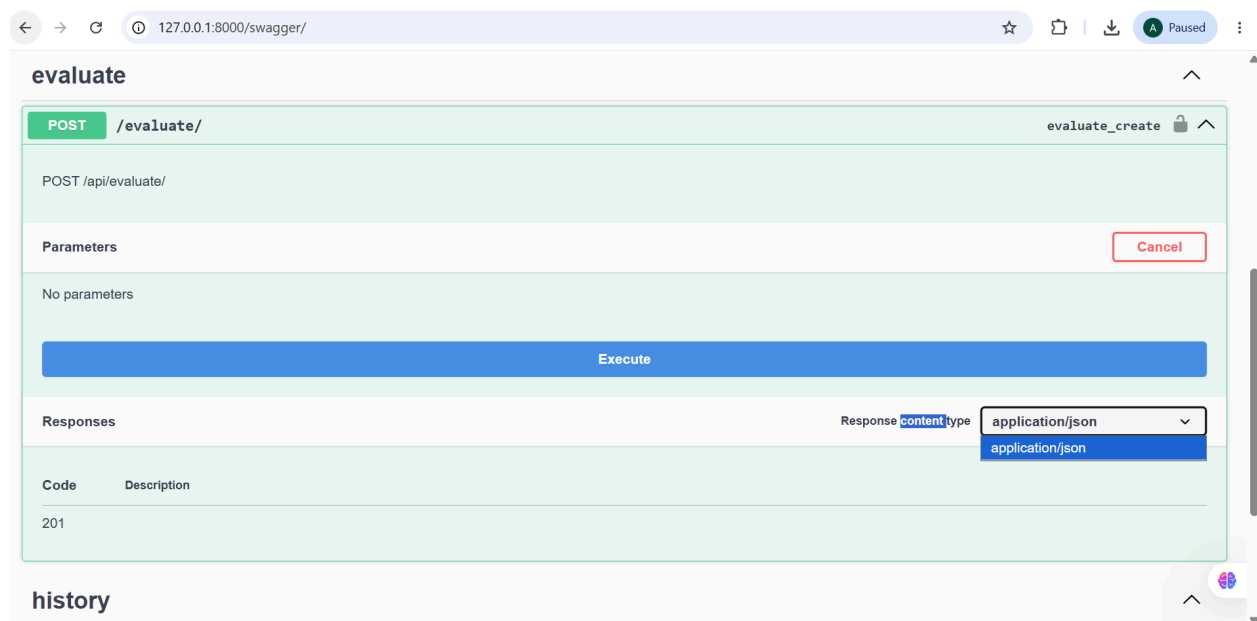
6. Django Async ORM Error

Issue:SynchronousOnlyOperation:You cannot call this from an async context

Cause:Django ORM queries were executed directly inside asynchronous functions.

Resolution:Database operations were moved outside async contexts or wrapped using asynchronous-safe techniques.

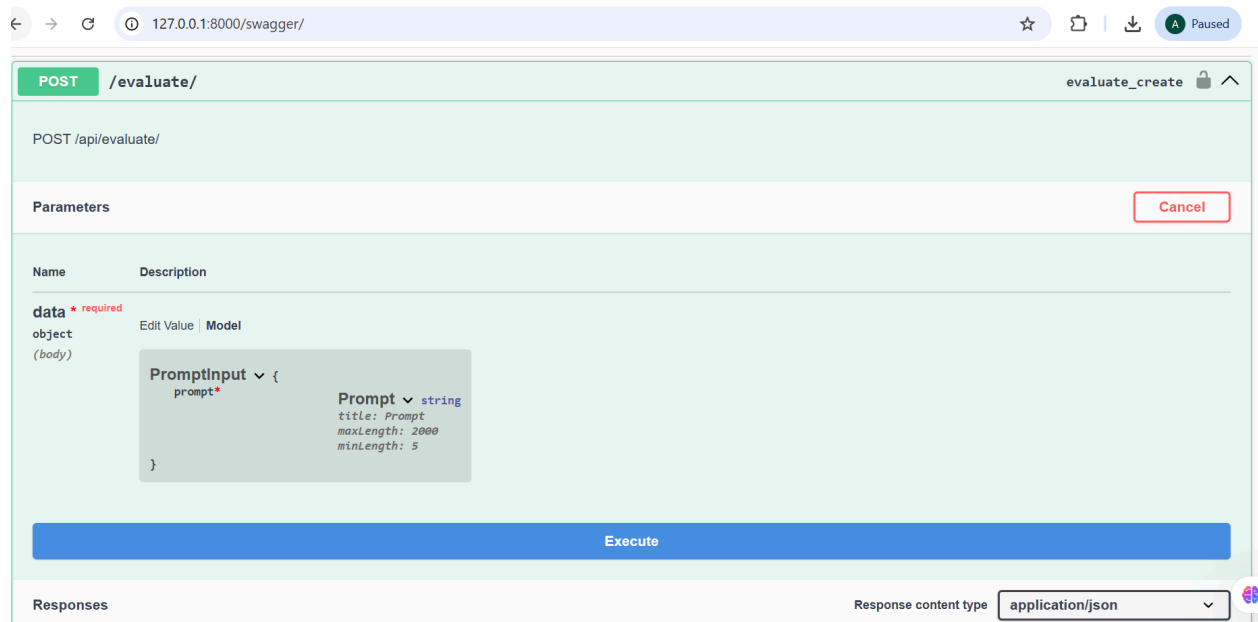
7. Swagger UI Schema Mapping Issue



Error: Swagger UI displayed "No parameters" for the `/api/evaluate/` POST endpoint.

Cause: The endpoint was powered by a standard Django REST Framework `APIView`. By default, `drf-yasg` (Swagger generator) cannot auto-detect the expected input data payload structure for raw `APIView` classes unless explicitly documented. Because there was no explicit link between the view and the data validator (`PromptInputSerializer`), Swagger rendered an empty parameter block.

Resolution: Decorated the view function with `@swagger_auto_schema(request_body=PromptInputSerializer)`. This explicitly mapped the serializer schema to the Swagger engine, turning the static documentation into an interactive JSON input portal with an "Edit Value" field.



8.huging face api did not worked

The screenshot shows a REST client interface with the URL `https://api-inference.huggingface.co/models/meta-llama/Llama-3.2-3B-Instruct/v1/chat/completions`. The request method is `POST` and the body is raw JSON:

```
{  "messages": [    {      "role": "user",      "content": "Explain Django ORM in simple words"    }  ]}
```

. The response section shows a 404 error: `getaddrinfo ENOTFOUND api-inference.huggingface.co`. A cartoon character is shown holding a sign that says "Could not send request".

Response

Could not send request

Error: getaddrinfo ENOTFOUND api-inference.huggingface.co

View in console • Learn about troubleshooting requests

getaddrinfo ENOTFOUND api-inference.huggingface.co — Could not send request.

9.aiml api did not work

The screenshot shows a REST client interface with the URL `https://api.aimlapi.com/v1/chat/completions`. The request method is `POST` and the body is raw JSON:

```
{  "messages": [    {      "role": "user",      "content": "Hello! Tell me a short joke."    }  ],  "temperature": 0.7}
```

. The response section shows a 404 error: `404 Not Found`. The response body is JSON:

```
{  "title": "Not Found",  "status": 404,  "message": "Model not found",  "instance": "/v1/chat/completions",  "requestId": "jGYz-f8jjenz5NBz-9qgm",  "timestamp": "2026-06-12T05:02:34.459Z",  "error": {    "name": "NotFoundException",    "message": "Model not found"  }}
```

Body

404 Not Found • 1.68 s • 982 B

JSON

```
{  "title": "Not Found",  "status": 404,  "message": "Model not found",  "instance": "/v1/chat/completions",  "requestId": "jGYz-f8jjenz5NBz-9qgm",  "timestamp": "2026-06-12T05:02:34.459Z",  "error": {    "name": "NotFoundException",    "message": "Model not found"  }}
```

404 Not Found — Model not found. NotFoundException at /v1/chat/completions.

Final Integrated Models

LLM Models (Response Generation)

Model	Provider	API	
deepseek/deepseek-v4-flash	Groq	api.groq.com	D
qwen/qwen3-32b	Groq	api.groq.com	A
nex-agi/nex-n2-pro:free	OpenRouter	openrouter.ai	B
gpt-oss-20b:free	OpenRouter	openrouter.ai	C

Judge Models (Evaluation)

Judge	Model Used	Scoring Criteria
Groq Judge	llama-3.1-8b-instant	Accuracy, Relevance, Completeness, Clarity, Conciseness
Gemini Judge	gemini-2.5-flash	Accuracy, Relevance, Completeness, Clarity, Conciseness

Both judges independently evaluate anonymized model responses (labeled A, B, C, D). Final score = Average(Groq Judge Score, Gemini Judge Score). Winner = highest adjusted score after consensus blending.

RESULT AFTER 30 EVALUATIONS

Gpt-oss-20b:free secures the highest wins as shown in the leaderboard .

```
[{ "model_name": "OpenRouter-gpt-oss-20b:free",  
  "total_wins": 17,  
  "average_score": 8.72,  
  "total_evaluations": 32 },  
{ "model_name": "OpenRouter-deepseek-v4-flash",  
  "total_wins": 8,
```

```
    "average_score": 8.28,  
    "total_evaluations": 30 },  
  {"model_name": "Groq-qwen/qwen3-32b",  
    "total_wins": 8,  
    "average_score": 8.26,  
    "total_evaluations": 34},  
  { "model_name": "Groq-llama-3.1-8b-instant",  
    "total_wins": 1,  
    "average_score": 7.8,  
    "total_evaluations": 5 },  
  {  "model_name": "OpenRouter-nex-n2-pro:free",  
    "total_wins": 1,  
    "average_score": 7.68,  
    "total_evaluations": 33}  
]
```